

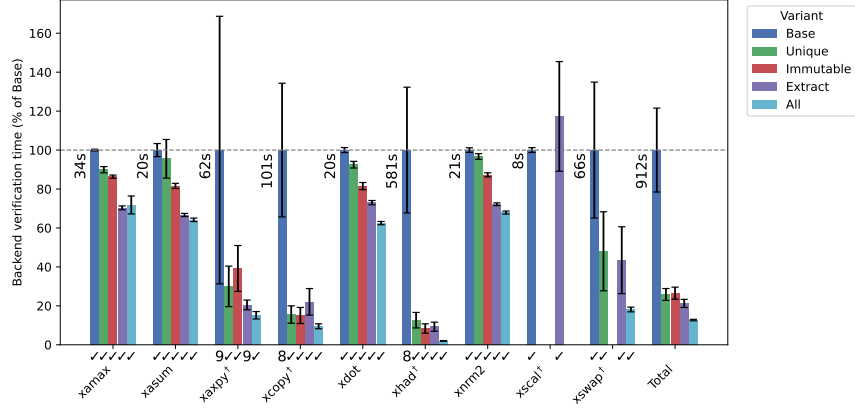
Kernel	# _{imm}	# _p	# _A	Result	Base		Unique		Immutable		Extract		All	
					#	T	#	T	#	T	#	T	#	T
xamax	1	4	31	✓	10	34	10	30	10	29	10	24	10	24
xasum	1	2	37	✓	10	20	10	19	10	16	10	13	10	13
xaxpy [†]	1	1	16	✓	9	66	10	19	10	24	9	13	10	9
				×	1	23	0		0		1	13		
xcopy [†]	1	1	15	✓	8	52	10	16	10	15	10	22	10	10
				×	2	296	0		0		0			
xdot	2	2	24	✓	10	20	10	18	10	16	10	15	10	12
xhad [†]	2	1	19	✓	8	450	10	73	10	48	10	54	10	12
				×	2	1104	0		0		0			
xnrm2	1	2	54	✓	10	21	10	20	10	18	10	15	10	14
xswap [†]	0	2	19	✓	10	66	10	32	2.1	-	10	28	10	12
xscal [†]	0	1	13	✓	10	8	-	-	-	-	10	10	-	-
Total					913		227	4.0	166	5.1	194	4.7	106	8.5

(a) CLBLAS level 1 kernels

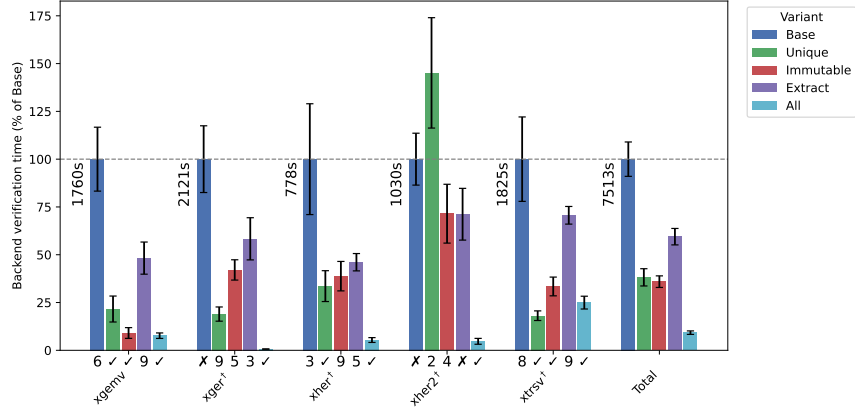
Kernel	# _{imm}	# _p	# _A	Result	Base		Unique		Immutable		Extract		All	
					#	T	#	T	#	T	#	T	#	T
xger [†]	2	3	69	✓	0	8 1751	9	369	5	981	3	1443	10	14
				×	8	1751	1	701	5	803	7	1149		
				T.O.	2	-	0		0		0			
xher [†]	1	3	37	✓	3	1507	10	261	9	280	5	401	10	42
				×	7	465	0		1	495	5	315		
xher2 [†]	2	5	56	✓	0	2 922	2	922	4	731	0	1.4	10	48
				×	10	1030	8	1638	6	739	10	733		
xtrsv [†]	2	3	68	✓	8	1381	10	330	10	610	9	1217	10	455
				×	0	0	0		0		1	1933		
				T.O.	2	-	0		0		0			
xgemv	2	3	80	✓	6	1664	10	380	10	159	9	787	10	135
				×	4	1905	0		0		1	1411		
Total					7514		2867	2.6	2699	2.8	4466	1.7	694	10.8

(b) CLBLAS level 2 kernels

Table 1: Verification results for the cBLAS GPU kernels for level 1 and 2. **Base** is the baseline configuration without any extra features. It is compared against versions with unique type qualifiers (**Unique**), immutable arrays (**Immutable**), extracted kernel bodies (**Extract**), and a configuration enabling all features simultaneously (**All**). The [†] indicates we (try to) prove functional correctness in addition to memory safety. #_{imm} and #_p is the number of immutable and normal pointer arrays, respectively. #_A the is the total number of annotations.



(a) Level 1



(b) Level 2

Figure 1: Verification backend times for the cBLAS GPU kernels for level 1 and 2, normalized so Base=100%. **Base** is the baseline configuration without any extra features. It is compared against versions with unique type qualifiers (**Unique**), immutable arrays (**Immutable**), extracted kernel bodies (**Extract**), and a configuration enabling all features simultaneously (**All**). Bars show mean with standard error of the mean. Below the bars: ✓ = 10/10 successful, ✗ = 0/10 successful, else number of successful runs out of 10. Next to base bars: average time in seconds of the 10 base runs. The † indicates we (try to) prove functional correctness in addition to memory safety.